# BROKEN CHAIN - NETWORK FORENSICS ANALYSIS

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## OBJECTIVE

Analyze a packet capture containing unencrypted HTTP traffic to identify exposed login credentials and analyze protocol usage.

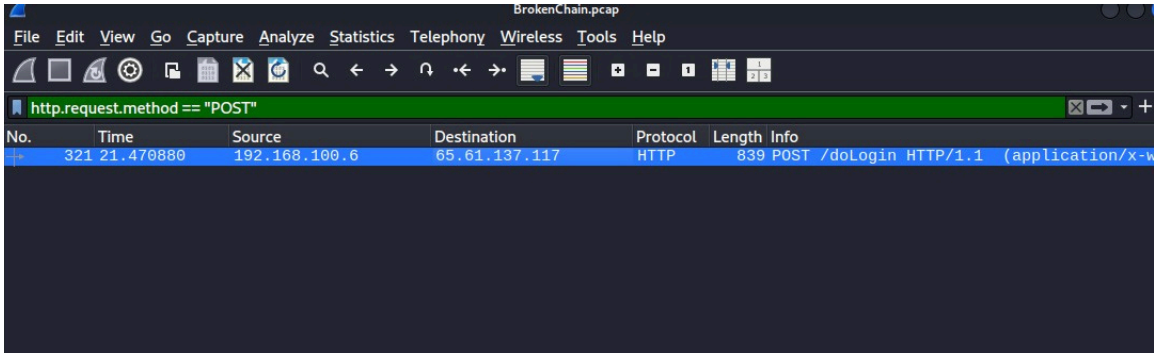
## CHALLENGE INFORMATION

Platform: **CTFROOM**
Category: **Network Traffic Analysis / Forensics**
Tools: **Wireshark**

## STEP 1: IDENTIFY THE LOGIN REQUEST

Filter Applied: http.request.method == "POST"

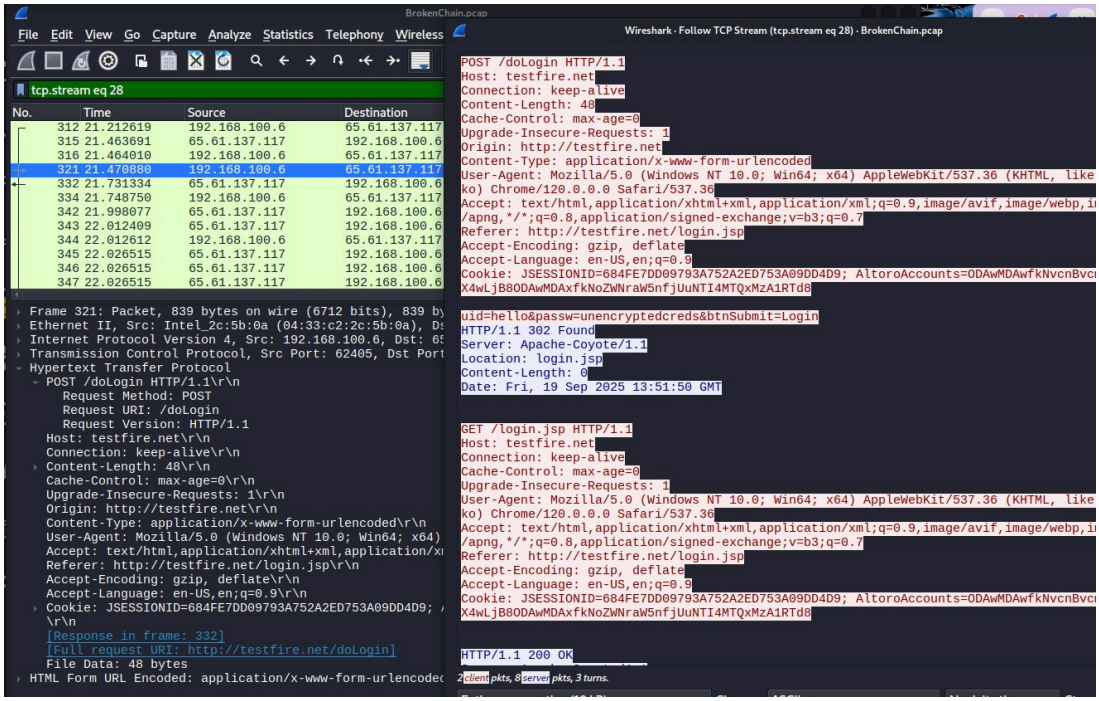
Result: Packet *#321-POSTrequestto/doLogin*




## STEP 2: EXTRACT CREDENTIALS

Method: *Right-clickpacket→Follow → TCP Stream*

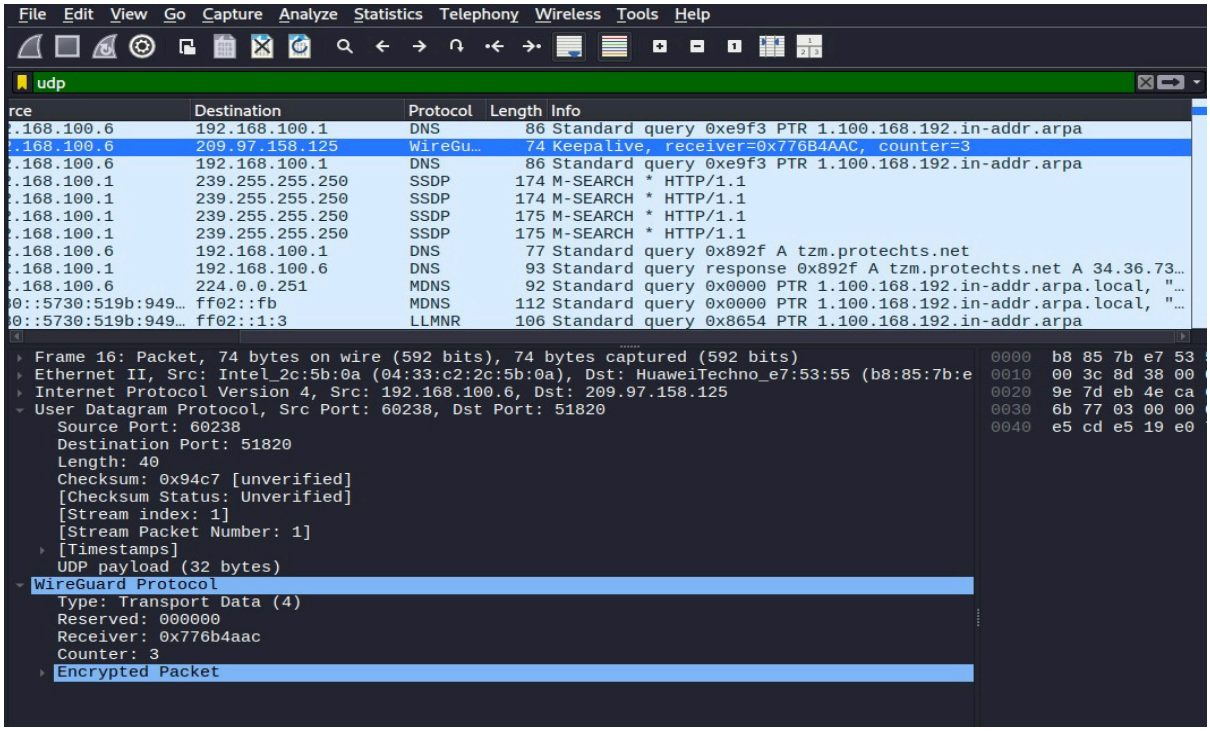
Credentials Found in Plain Text:
*uid=hello&passw=unencryptedcreds&btnSubmit=Login*




## STEP 4: IDENTIFY SECURE UDP PROTOCOL

Filter Applied: *udp*

Finding: *WireGuard VPN detected on port 51820*




**STEP 5: IDENTIFY SECURE TCP PROTOCOL**

Finding: *TLSencryptionpresentonport443(TLSv1.2, TLSv1.3)*

**SUMMARY OF FINDINGS**

Server IP: *65.61.137.117*
Domain: *testfire.net*
Login URL: *http://testfire.net/doLogin*
Username: *hello*
Password: *unencryptedcreds*
Insecure Port: *80*
Secure UDP Protocol: *wireguard*
Secure UDP Port: *51820*
Secure TCP Protocol: *tls*

TOOLS USED

*Wireshark 4.x*

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